

How a school swarms

collective motion for the aquarium level's ten followers · the Couzin zone model, measured against our own Forth implementation

Lengths are in **body lengths** (BL; a fish is 3.5 in). The tank is $13.4 \times 3.4 \times 4.7$ BL inside. A **school** is polarised (p_group near 1); a **swarm** is not (p near 0.1): moving one zone width, Δr_o , switches between them.

A · Four collective states from one rule, simulated here (N = 100, the paper's $r_r, \alpha, \theta, s, \sigma$)

A · swarm

top view

B · torus

spin-axis view

C · dynamic parallel

side view

D · highly parallel

side view

$\Delta r_o = 0, \Delta r_a = 15$
p_group 0.16 · m_group 0.10

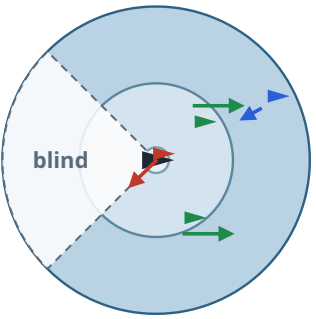
$\Delta r_o = 1, \Delta r_a = 12$
p_group 0.09 · m_group 0.90

$\Delta r_o = 3, \Delta r_a = 10$
p_group 0.90 · m_group 0.03

$\Delta r_o = 10, \Delta r_a = 10$
p_group 1.00 · m_group 0.02

ours · N = 100 simulated here; the zone widths are ours, the paper prints none

B · The model: three zones and a blind volume



repulsion zone, $r_r = 1$
anyone here: turn away, nothing else counts

orientation zone, width Δr_o
those here: match their heading

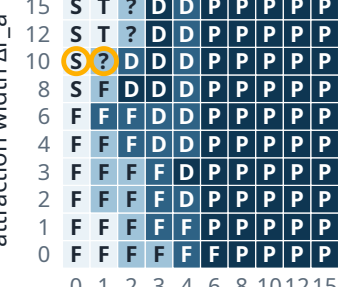
attraction zone, width Δr_a
those here: turn toward them

blind volume, 90° behind
 $\alpha = 270^\circ$: cannot see here

Drawn at the school setting: $\Delta r_o = 5, \Delta r_a = 6$

verified · zones, α, r_r : Couzin 2002**ours · widths drawn**

C · The map: p_group over the two zone widths



p_group (alignment)
0.25 0.6 0.9 0.9
S swarm: low p, low m
T torus: low p, high m
D dynamic parallel: high p
P highly parallel: p ≈ 1
F fragments (more than half)
? seeds disagree: no one state

0 1 2 3 4 6 8 10 12 15
orientation width Δr_o (BL)

ours · N = 100, 3 replicates, 700 steps (paper: 30)

D · The rule in eight lines, and the Forth word that does each

1 See the others, except behind

2 Repulsion if anyone is within r_r

3 Else orientation and attraction

4 Nobody in any zone: keep the heading

5 Turn toward d by at most $\theta\tau$, move s τ

6 The leader counts w times in sums 3

7 Walls repel like a neighbour

8 A dart startles the followers nearby

$u_{ij} \cdot v_i \geq \cos(\alpha/2)$
 $d = -\sum u_{ij}$ (nothing else counts)
 $d_o = \sum v_j, d_a = \sum u_{ij}; d = (d_o + d_a)/2$
 $d = v_i$
 $v \leftarrow \text{rotate}(v, d, \theta\tau); c \leftarrow c + s \tau v$
 $\sum w_j v_j, \sum w_j u_{ij}$
wall within 0.6 BL: one more repulsion
swim away from the leader for 0.5 s

sch-pair
sch-pair
sch-pair, sch-want
sch-want
sch-turn, sch-follow
sch-pair (weight)
sch-wall
sch-startle-all

The paper's random error ($\sigma = 0.05$ rad) is left out of the Forth: the caller adds it. Our units differ from the paper's, on purpose: at its $s = 3$ BL/s and $\theta = 40^\circ/s$ a 90° turn takes 6.75 BL of swimming (radius 4.3 BL), wider than the tank is deep (3.4 BL), and the tank runs below could not keep the fish in. Ours: $s = 2$ BL/s, $\theta = 120^\circ/s$, 1.50 BL, radius 0.95 BL.

verified · rules 1–5 and the paper's s, θ : Couzin 2002, eqns 1–3 and Fig. 3**ours · rules 6–8, our s, θ , the 0.6 BL wall, the 0.5 s startle**

F · The Forth core, school.fth: the four rule words, and what they cost

sch-pair	240 B · 15 lines	sch-want	201 B · 8 lines	sch-turn	283 B · 12 lines	sch-follow	211 B · 10 lines
one neighbour: blind volume, then repulsion, orientation or attraction : sch-pair dup 19 sc! 0 over sch@ 0 cur sch@ - 16 sc! 1 over sch@ 1 cur sch@ - 17 sc! 2 swap sch@ 2 cur sch@ - 18 sc! 16 vlen2 dup 0.000001 > over 1 par@ 0 par@ + 2 par@ + dup * < & if sch-sqrt dup 23 sc! 1 swap / 16 vscale heading-dot-u 3 par@ >= if 23 sc@ 0 par@ < if -1 0 acc+u 1 9 sc+! else 23 sc@ 0 par@ 1 par@ + < if 19 sc@ weight 3 acc+v 1 10 sc+! else 19 sc@ weight 6 acc+u 1 11 sc+! then then then else drop then ;	the wanted direction from the three sums, or the old heading : sch-want 12 vzero 9 sc@ 0 > if 0 vnorm drop 0 sc@ 12 sc! 1 sc@ 13 sc! 2 sc@ 14 sc! else 10 sc@ 0 > if 3 vnorm drop 3 sc@ 12 sc+! 4 sc@ 13 sc+! 5 sc@ 14 sc+! then 11 sc@ 0 > if 6 vnorm drop 6 sc@ 12 sc+! 7 sc@ 13 sc+! 8 sc@ 14 sc+! then then 12 vnorm 0.000001 > not if 3 cur sch@ 12 sc! 4 cur sch@ 13 sc! 5 cur sch@ 14 sc! then ;	turn toward it by at most $\theta\tau$, keep the heading unit length : sch-turn 3 cur sch@ 12 sc@ * 4 cur sch@ 13 sc@ * + 5 cur sch@ 14 sc@ * + dup 4 par@ >= if drop else >r 12 sc@ r@ 3 cur sch@ * - 20 sc! 13 sc@ r@ 4 cur sch@ * - 21 sc! 14 sc@ r> 5 cur sch@ * - 22 sc! 20 vnorm 0.000001 > not if 4 cur sch@ 20 sc! 3 cur sch@ negate 21 sc! 0 22 sc! 20 vnorm 0.000001 > not if 1 20 sc! 0 21 sc! then then 3 cur sch@ 4 par@ * 20 sc@ 5 par@ * + 12 sc! 4 cur sch@ 4 par@ * 21 sc@ 5 par@ * + 13 sc! 5 cur sch@ 4 par@ * 22 sc@ 5 par@ * + 14 sc! 12 vnorm drop then ;	one follower, start to finish: scan, walls, startle, turn, move : sch-follow 15 sc! 0 vzero 3 vzero 6 vzero 0 9 sc! 0 10 sc! 0 11 sc! sch-n 0 do i cur <> if i sch-pair then loop sch-walls sch-want sch-startle? if sch-startle-away sch-startle-tick then sch-turn 12 sc@ 9 cur sch! 13 sc@ 10 cur sch! 14 sc@ 11 cur sch! 0 cur sch@ 9 cur sch@ 6 par@ * + 6 cur sch! 1 cur sch@ 10 cur sch@ 6 par@ * + 7 cur sch! 2 cur sch@ 11 cur sch@ 6 par@ * + 8 cur sch! ;				

Dictionary bytes per word (zForth HERE), 2179 B in all

sch-turn	283
sch-pair	240
sch-follow	211
sch-want	201
sch-startle-all	123
28 other words	1121

2179 B

6.2 ms

1e-03

~0.6 ms

of the 64 KB dictionary (3.3 %); 79 code lines

one tick, 11 fish, on the Chromecast HD (median of 3 runs); x86 here 8.4 ms

largest heading error against couzin.py in one tick (the engine's sine is 0.2 % off)

per follower per tick: spread over frames, the full 10 Hz step is about 6 % of a core

ours · the Forth, its size, error and timing are our measurements: the interpreter only, not yet inside the engine

G · The Forth, running in the tank: swarm, torus, school, and a startle

swarm setting (0, 10)

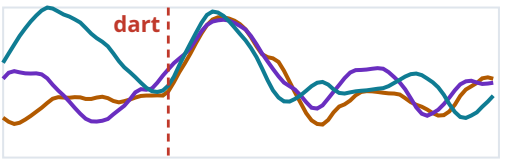
leader weight w
w = 1 2.7 BL · +0.02 · 0.27 · yes
w = 3 2.8 BL · -0.12 · 0.40 · no
w = 6 2.5 BL · +0.07 · 0.51 · no

torus setting (1, 10)

leader weight w
w = 1 3.4 BL · -0.05 · 0.58 · yes
w = 3 2.3 BL · +0.24 · 0.64 · no
w = 6 2.3 BL · +0.31 · 0.61 · no

school setting (5, 6)

leader weight w
w = 1 3.8 BL · +0.12 · 0.87 · yes
w = 3 3.5 BL · +0.17 · 0.84 · yes
w = 6 3.2 BL · +0.18 · 0.82 · yes

Startle: distance to the leader

swarm torus school
-6 s to +12 s; top = 6 BL; one seed

ours · the Forth core in the tank's real box (w = 3, seed 1 drawn; the table is the mean of 3 seeds; "no" = a fish strayed more than 0.5 BL out)

Sources · live links in the PDF; what could not be verified is marked

S1 [Couzin et al. 2002, J. Theor. Biol.](#)
verified zone model, states, metrics, Fig. 3

S2 [Couzin et al. 2005, Nature](#)
unverified informed minority guides a group

S3 [Pitcher 1983, via Wikipedia](#)
unverified shoal to school: a continuum

S4 [Ocellaris clownfish \(Florida Museum\)](#)
unverified site-attached: not schooling fish

S5 the aquarium level and this repository
ours tank, fish, zForth, our runs

How to read the chips
verified the source was opened; the number is on its page
unverified from a summary, or not opened
ours our own choice, maths or measurement
Not claimed: that real clownfish school (they do not), that our sweep shows the paper's hysteresis (it is too coarse), or that the Forth was timed inside the engine. Data dated 2026-10-01.